

Total No. of Questions : 4]

SEAT No. :

PD40

[Total No. of Pages : 2

[6409]-241

S.E. (AI & ML) (Insem)

OPERATING SYSTEMS

(2019 Pattern) (Semester-IV) (218552)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2 Q.3 or Q.4.*
- 2) *Assume suitable data, if necessary.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Figures to the right indicate full marks.*

- Q1)** a) Describe the differences between a monolithic kernel and a microkernel. **[5]**
- b) What is an operating system? List and explain services provided by the operating system. **[5]**
- c) Give the significance of following shell commands with an example: Is, uniq, tail, touch, grep. **[5]**

OR

- Q2)** a) What are the OS Design Considerations for multiprocessor and multicore architecture? **[5]**
- b) Describe with the help of a neat diagram the concept of simple batch systems. **[5]**
- c) Write a function in a shell script that calculates the factorial of a given number. **[5]**

- Q3)** a) What do you mean by threads? Explain the benefits of threads. **[7]**
- b) With the help of neat, explain in detail process state transition diagram with two suspend states. **[8]**

OR

P.T.O.

- Q4)** a) Discuss with the help of neat diagram circumstances under which CPU-scheduling decisions may take place. [7]
- b) Consider the Set of Processes with Arrival Time, Burst Time & Priority. [8]

Process	Arrival Time	Burst Time	Priority
P1	2	4	3
P2	4	3	2
P3	6	2	5
P4	8	8	1
P5	0	5	4

Find Average Turnaround Time & Average Waiting Time for Priority (Preemptive) & SJF (Non-Preemptive) scheduling algorithms with the help of Gantt chart.

